

one small radiomics cohort

$n_+ = 57$ events, $n_- = 126$, 1004 features

honest
nested CV

HONEST EVALUATION

0.47

$\approx$ chance
(leakage-free)

cherry-pick
best of splits

OPTIMISTIC EVALUATION

0.67

looks like signal
(split-shopping + leakage)

reported optimism

$$0.67 - 0.47 = 0.20$$



**the gap is sampling variance, predictable pre-data,
set by class counts, not dimensionality:**

$$\Delta \approx K \cdot \text{SE}_{\text{AUC}}(n_+, n_-)$$

validated across 50 radiomics cohorts:

closed-form SE

$$R^2 = 0.73$$

vs

dimensionality

$$R^2 = 0.09$$